

SMART KISAN MITRA: A UNIFIED WEB-BASED ECOSYSTEM FOR AGRICULTURAL EMPOWERMENT

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Abstract:

Agriculture is still an important way of life for people living in rural areas.. Farmers are having a hard time finding digital tools that work well together. Now they have to use many different apps for things like checking the weather looking at their crops and understanding market prices. This is very frustrating because it wastes their time uses up a lot of data and is especially difficult in areas with slow internet.

We have created Smart Kisan Mitra, an online platform that brings all these essential services together in one place. Our platform is built using an architecture with a Flask backend and a simple Single Page Application frontend. This makes it very lightweight and easy to use. It has features like real-time weather updates, prices of crops at the local market a tool to detect crop diseases using AI a guide to government schemes and a tracker, for expenses.

For checking crop problems we are using a type of computer learning called a Convolutional Neural Network, based on MobileNetV2. This is very accurate. Does not use too much processing power. The system also gives advice like when to water crops using simple rules. Our platform is very responsive because it uses APIs and loads data in the background. From our tests it seems that Smart Kisan Mitra uses much fewer resources and is easier to use than the usual methods. By bringing all these tools we hope to make it easier for farmers to make decisions and help more farmers use digital tools.

Keywords: Smart Agriculture, Single Page Application (SPA) Flask Framework, Crop Disease Detection. MobileNetV2, Convolutional Neural Networks (CNN), Mandi Price Analysis, Decision Support System.

I. INTRODUCTION

Farming is a part of the economy and the main way that millions of people make a living. Even though farming methods have improved a lot people in areas have not started using digital tools as much as they could. The main problems are not just that people do not have the equipment but also that the tools are not easy to use and people get tired of using so many different apps.

People have made apps to help farmers know what the weather will be like find out if their plants are sick and learn about market prices.. These tools are usually separate from each other. A farmer might use one app to check the weather another app to figure out what is wrong with a plant and a government website to check the prices of crops. Switching between all these tools is annoying and hard to do. Also many of these apps use a lot of data, which can be a problem for people with internet connections.

In addition, to taking care of crops keeping track of money is a problem that current apps do not solve. Farming is a business. Few platforms help farmers keep track of how much money they spend compared to how much money they make. It is also hard to find out about government help programs because the official websites are often hard to use.

Smart Kisan Mitra was made to solve these problems. We made a website that has all the tools that farmers need in one place. Because the website updates automatically without reloading the page it uses much less data. The system brings together weather updates, market trends, a "crop doctor" to diagnose plant problems a way to find out about government help programs and a way to keep track of money in one place. By making these tools easy to use we hope to help farmers use tools in their everyday work.

II. PROBLEM STATEMENT

Farming today needs information to make good decisions.. The computer systems that farmers use are not connected very well. To really know how their farm is doing and if it is making money farmers have to look at a lot of websites and programs. This causes a lot of problems: It is hard to use a lot of computer systems because they all work differently. This makes it confusing for farmers those who are not used to computers. When you have a lot of apps running on your phone or computer it uses up a lot of data and battery. This is a problem in areas where the internet connection is not very good. Different apps do not talk to each other. For example a weather app and a crop health app are separate. Farmers need to know how the weather will affect their crops like if a big storm's coming and their crops are sick. Most farming technology only looks at how food the farm is producing, not how much money the farm is making or spending. This is important for farmers to know so they can make decisions.

Smart Kisan Mitra is a system that solves these problems. It is one system that uses less data and is easy to use, even for people who are not good with computers. Smart Kisan Mitra connects information about the weather, crop health and the market so farmers can make decisions, about their farm.

III. LITERATURE SURVEY

Recent years have seen rapid development in isolated agricultural technologies. Convolutional Neural Networks (CNNs), for example, have become the standard for image-based crop disease detection. Research utilizing the PlantVillage dataset demonstrates that deep learning models—particularly lightweight architectures like MobileNetV2 [1, 2]—can achieve high accuracy with minimal computational overhead [3, 4, 5]. However, these models are usually deployed as single-function apps [6, 7]. They can identify a disease, but they do not provide the financial or weather context needed to treat it efficiently.

Similarly, weather forecasting APIs and market price platforms, such as Agmarknet, provide valuable raw data [8, 10]. Yet, they rarely offer tailored, farm-specific advisories. The data is presented, but the farmer must independently figure out what it means for their specific crop schedule.

Integrated platforms do exist, but they tend to be enterprise-level, resource-intensive, or commercially biased (steering farmers toward specific agrochemical products) [8]. Moreover, the financial management aspect of smallholder farming is noticeably absent from both academic literature and current app stores.

Our work builds on these individual advancements—such as IoT concepts and CNN-based diagnostics [6, 9]—but shifts the focus toward integration and lightweight deployment, creating a unified ecosystem rather than just another isolated tool.

PROPOSED METHODOLOGY

The Smart Kisan Mitra system is made to help farmers deal with the problems of using different agricultural applications relying too much on data and not having the right tools to make good decisions. The Smart Kisan Mitra system is a simple and easy to use website that brings everything together. It is not like agricultural websites that only do one thing. The Smart Kisan Mitra system makes sure that all the different parts work together. The Smart Kisan Mitra system is designed to be fast and easy to use. It gives farmers the information they need to make decisions. The Smart Kisan Mitra system does this by breaking down the work into steps.

The Smart Kisan Mitra system does not use a lot of complicated programs. It uses a system where each part does one job and works well with the other parts. Each part of the Smart Kisan Mitra system talks to the others, through a system. The work that the Smart Kisan Mitra system does is divided into steps. Each step does one thing to help with data.



Step 1 – Unified Data Acquisition and Input Handling

The Smart Kisan Mitra system has a goal to make it easy for people to give it information. Many systems now have a problem. They do not have a simple way for people to give them information. People have to fill out a lot of forms and click on buttons. This makes it hard for people to use these systems, people who live in rural areas.

The Smart Kisan Mitra system is trying to fix this problem. It has a way for people to give it information. The system has a page where people can do everything they need to do. They can select what kind of crop they have, how money they spent and even upload pictures. All this information is sent to the

system in a way that's easy for it to understand. For example if a person wants to know if their crop is sick they can upload a picture of it. The system will then look at the picture. Tell them if their crop is sick or not. This way people do not have to fill out a lot of forms or click on buttons.

Step 2 – Data Structuring and Preprocessing

When the system gets the information from the person it has to make sure it is correct and in the format. This is an important step because it helps the system work well and give good results. The system checks the information to make sure it is complete and correct. For example if a person is telling the system how money they spent it checks to make sure the number is correct and not negative. The system also checks to make sure the picture of the crop is in the format. If the information is not correct the system will tell the person. Ask them to fix it. This helps the system work well and give results.

Step 3 – Modular Processing and Logical Execution

After the system has checked the information it sends it to the part of the system to be processed. The system has different parts and each part does something different. The weather part of the system gets information from sources and uses it to tell people what the weather will be like. The price part of the system looks at the prices of crops and tells people what they should do. The disease part of the system looks at pictures of crops. Tells people if they are sick or not. Each part of the system works independently which means they do not interfere with each other. This makes the system work well. Gives good results.

Step 4 – Decision Support and Insight Generation

The Smart Kisan Mitra system is not a system that gives people information. It is a system that helps people make decisions. It takes all the information it has. Uses it to tell people what they should do. For example the weather part of the system does not just tell people what the weather will be like. It tells them what they should do based on the weather. The price part of the system does not just tell people the prices of crops. It tells them when they should sell their crops. This way the system is not just giving people information. It is helping them make decisions. Do what is best for them. Asynchronous Communication and Response Handling The system is designed to work and not be slow. When a person asks the system for something it does not make them wait. It sends the request to the part of the system and then waits for the answer.

When the system gets the answer it sends it back to the person. Updates the page. This way the person does not have to wait for the page to reload. This makes the system work well and be fast.

Step 5 – Dynamic Output Rendering and Visualization

When the system has the answer it shows it to the person in a way that's easy to understand. It uses pictures and charts to help people understand the information.

For example the expense part of the system shows people how money they have spent in a chart. This makes it easy for people to see where their money is going. The system is designed to be easy to use and understand. It shows people the information they need in a way that's clear and easy to understand.

Step 6 – Performance Optimization and Scalability

The final step is to make sure the system works well and can handle people using it. The system is designed to be fast and efficient.

It uses a framework and does not have many dependencies. This makes it work well and be fast. The system is also designed to work even when the internet is slow. The system can also be expanded to do things. New features can be added without making the system slow. This makes the system a good choice for people who need a system that can help them make decisions and do what is best for them. The Smart Kisan Mitra system is a choice, for people who live in rural areas and need a system that can help them with their crops and money.

CONCLUSION

The **Smart Kisan Mitra** system fixes a problem in digital farming. It brings together services into one system. This makes it easier for farmers to use and access.

The system works well in areas with poor internet. It uses a web-based application built with basic JavaScript. This reduces data usage by cutting out framework load and frequent page reloads.

The frontend works with a backend built with Flask. It manages modules, such as:

- Weather advisories that help farmers plan
- **Smart Kisan Mitra** tracks market prices
- Information on government schemes
- Expense tracking for farmers

The system is designed to be -proof. It can support technologies like **MobileNetV2** for detecting crop diseases.

The system works with low delay and is easy to use. However it currently uses market data and lacks storage for community data.

The **Smart Kisan Mitra** project bridges the gap between technology and practical use. It provides a foundation for technology-driven change, in agriculture. The **Smart Kisan Mitra** system helps farmers. The **Smart Kisan Mitra** system makes farming.

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